

e-NAM's Cross-Market Price Impact: Evidence from India's Wheat Mandis

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For decades, the mandi system has served as a cornerstone of agricultural marketing in India. These physical marketplaces, managed by Agricultural Produce Market Committees (APMCs), connect farmers to buyers, ensuring a steady flow of agricultural products to consumers. However, concerns have been raised about the mandi system's struggle to integrate farmers effectively into the broader network of wholesalers and traders. This lack of integration limits market reach and can perpetuate unfair pricing practices. This paper delves into the concept of market integration and explores how e-NAM, an electronic trading platform, can potentially address these issues. By bypassing the limitations of the traditional mandi system, e-NAM can offer a pathway toward greater price stability and supply chain efficiency for farmers.

This paper investigates whether and to what extent the enrollment of APMCs in the e-NAM platform influences price convergence for wheat across Indian markets. We analyze the impact of e-NAM enrollment on two distinct specifications of price convergence during the period 2003 to 2019.

To assess the causal impact of e-NAM's 3-period enrollment on wheat price convergence, we employ an event study approach. We define e-NAM enrollment as the treatment variable, and our outcome utilizes two standardized price convergence specifications: within-APMC prices (benchmarked at the APMC average) and across-APMC prices (benchmarked at the state average). Initially, we control for the proportion of previously enrolled APMCs per state, the credit-to-deposit ratio, and the standard of living to account for potential selection bias. Later, we incorporate other covariates such as mandi-level wheat suitability and a mandi's proximity to the district border.

Contrary to expectations, e-NAM enrollment is associated with a 0.6 deviation below the APMC price average for mandis within enrolled APMCs and there's no significant impact on price convergence across APMCs. These observations remain consistent after incorporating covariates for wheat quality variation (given our suspicion of a two-tiered market) and conducting tests to observe heterogeneity across waves, alternative benchmarks, and any endogeneity of nonresponse.

This paper concerns the field of literature combining spatial analysis with market integration. While existing research focuses on price discovery at the state level or across commodities, we analyze the price dispersion across mandi and APMC boundaries. This allows us to leverage the minute arbitrage opportunities presented to intermediaries within the mandi system — nuances like transport costs and quality variation. Further, amongst vast literature on monsoon crops like rice and maize, we investigate the incredibly perishable and temperature-dependent winter crop, wheat.

The rest of the paper is organized as follows. Section 1 provides an overview of India's mandi system and e-NAM, and discusses the opportunities and shortcomings presented to producers and

middlemen. Section 2 describes our data and variable construction, empirical framework, and sample. Section 3 presents the results of our baseline estimates and additional covariates. Section 4 presents our validity concerns and the result of our robustness checks, and Section 5 concludes.

1 India’s Agricultural Setting

Two key policies at the forefront of India’s agricultural price regulation are the Agriculture Produce Marketing Committee (APMC) Acts and the National Agriculture Market (e-NAM) platform. In the absence of a competitive market structure, the APMC Acts were rolled out during the 1960s and 70s, through which the geographical area of participating states was divided into smaller market areas, also called “mandis,” and became managed by a Market Committee regulated by state governments. APMCs’ notable principles include setting a single market levy across within-state sales, preventing wholesale marketing activities, and prohibiting direct sales between retailers and farmers such that produce has to be sold to a licensed intermediary. While aiming to protect farmers during the decades of Indian self-determination after India’s independence in 1947, the APMC system is arguably outdated. APMCs may restrict market access by creating smaller, regulated mandis, limiting competition, and potentially allowing dominant players to collude by leveraging access to information. This can lead to artificially inflated prices for consumers and lower prices for farmers. While the APMC institutes a minimum support price to prevent purchases below a certain threshold, the lack of unification among sellers and intermediaries creates an environment ripe for exploitative practices and limited information access. On the producer side, some of the major problems of this mandi system include the insufficient number of APMC markets and their inadequate infrastructure, the barriers of entry for small farmers, the low remuneration to farms and the high intermediation costs, the lack of market information, the need to *physically* bring produce to the mandis, and the large market fees and licenses for trading. There is also a large risk of entering highly exploitative alternative channels of trading, whether the bidding process is largely opaque.

As a direct response to the limitations and pitfalls of the existing APMC system, e-NAM launched in 2016 as an online platform where farmers can compare prices across various markets and choose the best option, reducing their reliance on intermediaries and information asymmetry. Notably, this platform is meant to leverage the existing physical infrastructure of the mandis rather than being a parallel marketing structure to the APMC system. As a farmer, the gains of joining e-NAM are best described by the bidding process described below. With real-time data on bids across different markets, there is hope that transparency exposes local traders to competing offers, making it harder for them to manipulate prices downwards.

The conceptual framework of this bidding process is depicted in Figure 1, which also illustrates a simplified flow of payment, information, and goods under the network. Farmers, Farmer Producer Organizations (FPOs), and even commission agents can act as sellers on the platform, listing their products for bidding. Meanwhile, existing intermediaries like traders and potentially even retailers can participate as buyers alongside the local middlemen at the mandi. At the core of the system is the e-NAM portal, where sellers list their produce and buyers submit bids. This online platform facilitates the matching process contingent on the following events: farmers upload the minimum price for their lot onto the portal; bidders must complete the physical verification of the produce before offering a price which is uploaded onto the portal; after bidding according to these guidelines, the farmer will accept the bid and the buyer will start the payment process. While e-NAM handles the online listing and matching, physical aspects of the transaction, like delivery and quality checks,

still depend on the APMC infrastructure. That is, APMCs continue to facilitate this trade via amenities such as e-NAM's assaying facilities provided by the Government of India and accredited warehouses instituted by the Warehousing Development and Regulatory Authority of India. This allows farmers to preserve the quality of their produce and engage in futures contracts, which necessitates access to credit and storage. Once a bid is accepted, a virtual contract is established between the buyer and seller. The portal then generates a sales voucher that enables the buyer to pay directly to the bank account of the farmer. This process is mediated by commission agents of the APMCs, ensuring that there is no default on behalf of the buyers. Information flow is a crucial aspect of e-NAM. The platform provides real-time bid information from various buyers, offering farmers greater price transparency. It might also disseminate market trends, mandi arrivals, and government regulations to keep participants informed [17].

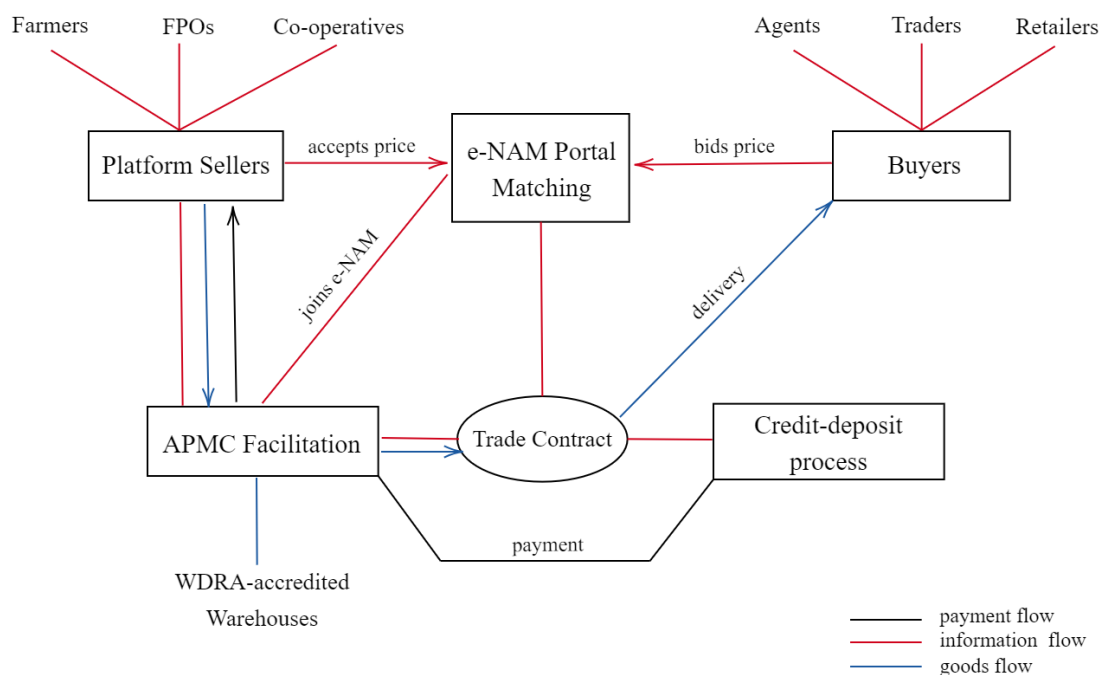


Figure 1: Simplified e-NAM trade process

With the goal of covering 585 APMC mandis across India, the e-NAM project instituted 3 schedules of adoption: 250 mandis in April 2016, 200 mandis in March 2017, and 135 mandis in March 2018. This staggered schedule was largely to accommodate the mandi-level reforms needed. While there are no state-wise restrictions to joining, APMCs would have to individually incorporate an additional provision for e-auction/trading as a mode of price discovery, and then abide by a single market fee and a single trading license to be valid across the state [23].

1.1 Farmer Side

Farmers engaging in the traditional mandi system have to face considerable uncertainty: poor storage infrastructure for futures contracts, hikes in inputs due to climate causes, and a lack of price assurance in the form of late payments and unjustified price reductions.

In light of this, agriculture policies in India have focused on three main tools to increase and stabilize the income of farmers as they use mandis: (1) improving crop yields with better farming practices, (2) decreasing input costs via subsidies, and, most relevant to this paper, (3) increasing output prices while stabilizing incomes through the Minimum Support Price (MSP). Despite these policies geared at improving farmer welfare, mandis continue to be unattractive for farmers as these policies fail to incorporate smallholder farmers into the broader network of wholesalers.

In this section, we delve into the literature on market integration and how achieving it via e-NAM, for instance, is the only means to bypassing these regulatory issues and achieving price and supply stability for farmers.

1.1.1 The Law of One Price

This section establishes the rationale behind focusing on price convergence as a vehicle for examining market efficiency. The law of one price serves as a critical foundation for this analysis. According to the law of one price, when given identical or highly substitutable goods, like wheat, each good has a single equilibrium price across the markets in the country [16]. If this law holds, such a market would be characterized as an "integrated market." Market integration, for the purposes of this paper and according to the literature [16, 3], can be defined as the level of price transmission — the phenomenon of upstream price changes impacting downstream prices — between two related markets. While upstream price changes can be changes in inputs or processing facilities, we are particularly concerned with the price change that arises when producers are quoted a different price than the market value of their produce. Interestingly, increasing distance has been found to reduce the speed of price transmission in domestic markets — that is, the gap in [16].

The law of one price also assumes an efficient market with perfect competition. In such a market, any price discrepancies for identical goods will be short-lived due to arbitrage opportunities. Those who engage in arbitrage would exploit price differences for profit, buy wheat in a cheaper market and sell it in a more expensive one, driving prices closer together until the arbitrage opportunity disappears. Further, if two markets are integrated, it is profitable for producers to sell in the market where goods are more expensive. The movement will continue until prices equate across markets. Even geographically separated markets can be spatially integrated if goods and information flow freely among them and, as a result, the effects of price changes in one market are transmitted to another market's price.

Ultimately, the conjecture is that, with no more price discrepancies, the price of these homogeneous goods will amount to no more than the costs of trade between two locations. Notable trade costs include transportation, insurance and hedging, and unobserved costs like searching for traders and information. While the law of one price seems to hold under stringent conditions like product homogeneity and no exchange rate, in the context of intra-country trade, this law seems more achievable if we reduce trade costs such as transportation and searching.

1.2 Intermediary Side

In an attempt to make mandis more attractive, intermediaries also provide a number of services, three of which are to hold inventory, reduce asymmetric information by using their reputation, and gather, organize, and evaluate information that is dispersed in society. Further, they utilize their network to provide linkages between credit and marketing in the absence of farmer-friendly institutions (i.e. banks that do not offer agricultural credit) [8].

Despite these services to equalize the playing field between farmers and wholesalers, power asymmetry is still maintained. Firstly, larger traders are not incentivized to directly trade with smallholder farmers due to low volumes and high search and transaction costs. They might also avoid engaging in the poor price-quality linkage that occurs without a formal assessment of produce quality in the mandis. Thus, direct purchases are dominated by smaller intermediaries. As a result, in the long run, these policies and the broader APMC system become more beneficial to the intermediaries than the farmers. In fact, there is an ongoing issue of imperfect competition with many farmers and a few middlemen. The most significant consequence of this is the collusion among middlemen that depresses the prices realized by farmers. This begs the question of whether alternatives exist to make the markets more welcoming to smallholder farmers?

We argue that introducing an electronic trading platform like e-NAM would improve the prices realized by farmers as it would increase competition in markets and decrease the collusion done by intermediaries. This follows Schmitz’s (2020) conjecture called the “Threatened Intermediaries Hypothesis,” which claims that intermediaries compete with e-commerce avenues to provide methods of trade that reduce transaction costs [26].

1.3 Existing Literature

The agricultural marketing system of India is characterized by various shortcomings, some of which are discussed above. However, more discussion needs to be had about the producer and intermediary perspective.

The recent integration of information technology within the mandi system has been approached from a broad state-level perspective or using a basket of commodities for analysis. For instance, there exists pre-e-NAM literature on the price discovery and the process of e-tendering adopted by certain APMCs in collaboration with the National Commodity and Derivatives Exchange (NCDEX), which by construction restricts its participating trading commodities to non-perishables. These studies mainly focused on the producer-side arbitrage where the farmers commanded the prevailing price rather than the intermediaries [18]. Further, several studies have detailed the beneficial effect of information and communication technology on agricultural marketing and supply chain efficiency in India [31, 8, 30].

Understanding that there is a lack of literature on the potential causal impact e-NAM has on mandi dynamics, this paper embarks on the following intended contributions to the literature on e-NAM, wheat mandis, and their relation to the price discovery process.

Thus, the first intended contribution of this study is to establish a causal channel between mandis joining e-NAM and mandis experiencing greater price stability proxied by price convergence (discussed below). Recent studies on the price discovery mechanism underlying e-NAM have focused on the differences between e-NAM futures prices and India’s largest agricultural derivatives exchange, the NCDEX [24, 13]. Manogna and Mishra (2021) have documented general price discovery across spot, futures, and mandi markets for nine specific commodities traded on the NCDEX prior to the surge in e-NAM literature [24]. As an extension of that study, Garg et al. (2023)

acknowledge e-NAM as a new spot price series only in the context of its dynamics between futures prices and established spot markets like NCDEX [13]. This broad focus on commodity-level price analyses fails to account for variation attributed to the particular mandis trading these commodities. With each mandi bearing a particular suitability for a given commodity, a particular price regulation imposed by its state, and a particular accessibility to infrastructure given its geography, there is ample space in the literature for a spatial analysis of the movement of price due to e-NAM participation.

The second intended contribution is to a broader discussion of cross-mandi information asymmetry and the possible arbitrage benefits of e-NAM. Numerous studies indicate that agricultural markets are often characterized by information asymmetry, where intermediaries possess superior knowledge about market conditions compared to farmers. This asymmetry can disadvantage farmers, as intermediaries can potentially negotiate lower prices based on their information advantage [2]. Limited rural connectivity, by restricting access to real-time market information, can exacerbate this information asymmetry. Recent research highlights the influence of digital platforms on consumer decision-making. Anderson and Magruder (2012) demonstrate how online review platforms like Yelp can significantly impact restaurant reservation availability, suggesting consumers rely on these reviews to form beliefs about product quality [5]. While this study focuses on the consumer perspective, it underscores the potential for electronic platforms to influence market dynamics through information dissemination. While Anderson and Magruder investigate how digital platforms influence consumer decision-making, Aker and Fafchamps (2015) narrow the focus to mobile platforms and information asymmetry in agricultural markets [4]. Aker and Fafchamps examine the impact of mobile phone coverage on price dispersion for agricultural products in West Africa. Their findings show that improved mobile phone coverage can reduce price dispersion for perishable goods like cowpea, but not necessarily for staple grains like millet and sorghum. The authors suggest that storage capacity might influence this effect, as farmers with storage have more flexibility to wait for better prices. Thus, existing literature provides valuable insights into information access, price discovery, and the role of technology in agricultural markets. However, there is a scarcity of research exploring the intermediary perspective in the context of rural connectivity and price arbitrage opportunities for producers. This paper will partly explore how information asymmetry between markets in e-NAM and not in e-NAM allows intermediaries to gouge prices vis-à-vis farmers in the Indian wheat market.

The third intended contribution is the nuanced effect of e-NAM on perishable goods in general by using the Rabi (winter) crop wheat amidst the widespread literature on Kharif (monsoon) crops such as rice and maize. One such Kharif-focused study is by Kaur et al. (2023), which identifies several limitations associated with alternative marketing channels like e-NAM and the electronic negotiable warehouse receipt (e-NWR) [15]. Their findings highlight limited e-NAM adoption amongst maize producers, often due to factors like the requirement for physical transportation to mandis and lack of credit access through e-NWRs. Given these limitations with alternative channels, futures markets emerged as a more viable option for their study's focus on hedging price risk in maize production. As aforementioned, this focus on futures markets, especially if it is due to a limited utilization of e-NAM, underscores the need for further research on the incentives underlying e-NAM participation. Considering those limitations for maize, our study attempts to study a different commodity with greater e-NAM coverage — wheat.

2 Data

This paper investigates whether and to what extent the enrollment of APMCs in the e-NAM platform influences price convergence for wheat across Indian markets. We analyze the impact of e-NAM enrollment on two distinct specifications of price convergence from 2003 to 2019.

This section first describes the reasoning behind the specifications of price convergence and the covariates utilized for identification. This is followed by an overview of data collection process and our empirical strategy.

2.1 Capturing "Price Convergence"

We draw inspiration from the economic concept of the law of one price, which assumes a frictionless market with perfect information and identical goods. We leverage the concept of relative price convergence, acknowledging that perfect price convergence may not be achievable due to factors like transportation costs or local variations in quality. However, we aim to capture the extent to which e-NAM facilitates price movement towards a single, more unified market value.

We will formally define a notion of price convergence that, for our purposes, captures the persistence of the gap between the income realized by farmers and the market value of their produce. Rajesh Kumar in *Strategies of Banks and Other Financial Institutions* (2014) describes two processes of price convergence: absolute and relative. The former, absolute price convergence, is when "relative prices" — either belonging to producers of close spatial proximity or producers with goods similar in product variety — converge to a zero gap. The latter, relative price convergence, is more lenient, allowing these relative prices to converge to a non-zero gap [20].

Notably, the relative price model presented by Alfredo Garcia-Hiernaux et al. (2023) illustrates a spatial component of price convergence. In their work analyzing price movement in Europe after the conception of the Euro, they report empirical evidence of there being a lateral convergence in prices; that is, there is relative price convergence between neighboring countries such as Germany, France, and Italy or Spain and Portugal [12]. They attribute this convergence to a spatial "co-integration" of prices, where prices increase or decrease together in the long run because they are influenced by the same fundamental forces (often geographic ones). While Alfredo Garcia-Hiernaux et al. extensively discussed the dynamic of being "neighbors" and sharing a geographic border, Chatterjee (2017), in his analysis of spatial competition and the potential behind a National Agricultural Market, focused more on in-state variation in competition than out-of-state [10]. Chatterjee cites evidence that, firstly, between-market competition increases the prices farmers get for their output and, secondly, competition in neighboring states does not influence those prices. The mechanism for this between-market price increase depends on an upward price convergence as a result of better price discovery. The hypothesis is that a farmer is likely to be offered a higher price given the combination of the farmer's access to non-local traders and their negotiations with the regular traders at their market site. While this reasoning extends to e-NAM applications, where these negotiations occur digitally with a broader network of traders, the validity of Chatterjee's second claim of negligible cross-border impacts on prices is threatened by e-NAM's transparency of out-of-border market transactions. That is, both farmers and traders on the platform are aware of the market competition elsewhere and can engage in competitive arbitrage to influence prices. Thus, this paper will attempt to address whether e-NAM's digital network increases the effect e-NAM participation has on price convergence within and across APMCs. Further, despite the convincing spatial, between-market argument, this still begs the question of whether observing "relative price

convergence” reflects fundamental information and whether it is equal to the fundamental value of the good. Our pursuit for *informative* price convergence should, then, combine the concept of relative price convergence and price discovery, which is essentially the way markets respond to new information.

Interestingly, the literature connecting e-NAM and price discovery is often either promotional [19, 11, 32], focused on promoting household adoption, or a discussion of futures and spot price differentiation across a vast basket of commodities [24]. The former outlook of price discovery (predicated on the adoption of some new technology) involves factors that are difficult to observe, such as technology’s ”perceived ease of use,” potential gains to social influence, and some measure of trust. Chaudhary and Suri (2021), in particular, attempt to model e-NAM’s adoption by simulating its decentralized appeal through structures such as blockchain [11]. In light of this lack of observable predictors of technological adoption, the piloting of e-NAM offers a natural experiment to look at real-time price movement. To analyze e-NAM in contrast with the literature on the older monopsony model of APMCs, we will define price convergence using two specifications that echo between-market and out-of-state effects of joining e-NAM.

The first specification measures the monthly price differential across mandis in the same APMC, benchmarked at the APMC’s average monthly price. This will test Chatterjee’s emphasis on between-market competition, where we observe whether mandis belonging to an APMC in e-NAM experience relative price convergence towards the APMC average, likely due to increased arbitrage. Not only does this specification allow us to control for the unique provisions per APMC (e.g., minimum support prices), but it also allows us to quantify the direct impact of e-NAM on the APMC, rather than the spillover effects it may have on neighboring APMCs. We consider this impact direct because if an APMC joins e-NAM, we know all its mandis obtain access to the platform.

The second specification measures the monthly price differential across APMCs, benchmarked at the state average monthly price. Contrary to the first specification, this is able to quantify the indirect impact of e-NAM on the price convergence across APMCs in the same state. We use the state average as a benchmark since agriculture in India is under the purview of the state. However, we conduct a robustness check benchmarking prices at the national wholesale price index for wheat. Further, we describe this impact as indirect since APMCs are individually involved with e-NAM rather than cooperating as a state. Thus, if convergence across APMC prices is seen after the implementation of e-NAM, it is likely that APMCs that are not in e-NAM are benefiting from the competitive forces underlying e-NAM.

2.2 Uniqueness of Wheat

As we are only concerned with markets within India, we ignore the typical criticism of the law of one price, which is that being able to capitalize on competitive markets across borders depends on exchange rates [7, 22]. However, there is still concern about how we can model the commodity arbitrage that predicates the law’s claim that prices converge at the point where no more arbitrage can occur.

For instance, given the multiple e-NAM periods, we might fail to capture the speed of price transmission, where price transmission may be incomplete in the short run but complete in the long run. In this case, ”complete” price transmission is a situation where, given two spatially separated markets, a price change in one is instantaneously transmitted to the other price while ”incomplete” price transmission is lagged [25]. If we do encounter incomplete short-run price adjustments, it would be difficult to test whether it reflects no price transmission at all or just an initial shock

to prices due to e-NAM’s earlier waves. Hence, we aim to pursue a commodity market that has empirical evidence of displaying price transmission.

Our selection of wheat mandis is motivated partly by Ghosh and Ghoshray’s (2011) study on whether India has integrated wheat markets. Analyzing pairs of wheat prices between regions within states, they discover a long-run relationship between all pairs of wheat prices in the absence of region-specific information [14]. This allows them to attribute variation across market pairs to factors like the institutional structures of the markets. With this evidence that India’s wheat markets have features of being an integrated market, we can hypothesize that wheat prices across APMCs would display some long-run price transmission. Thus, if arbitrage were to occur on the e-NAM platform, the literature supports an eventual plateau in arbitrage activity and a convergence in prices.

Further, as seen in Figure 2, the wholesale price inflation of wheat compares favorably with other grains with the exception of Ragi and other outliers. Thus, our observations of the wheat mandis can be comparable to the dynamics of other grain commodity prices.

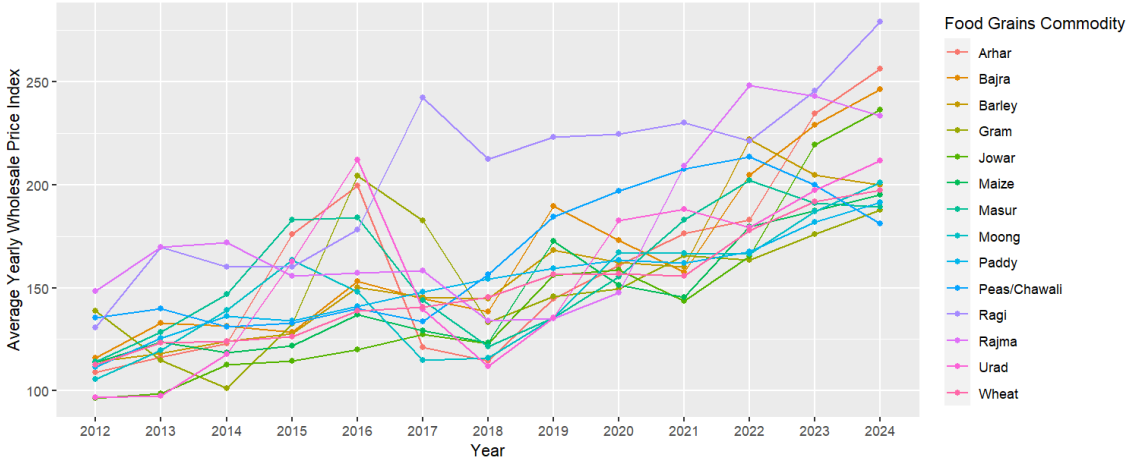


Figure 2: Wholesale price inflation for wheat and other food grains

2.3 Confounding Incentives to Joining e-NAM

To strengthen the validity of our findings on e-NAM’s impact, we need to consider potential biases that might influence which APMCs choose to join the platform. Here, we address three anticipated factors of selection bias: the participation of neighboring APMCs, the pre-e-NAM dependence of intermediaries, and the household standard of living.

There might be a selection bias based on heterogeneous priors. This means APMCs might be influenced by their expectations of e-NAM, shaped by how it’s impacting neighboring APMCs that have already joined. For example, if neighboring APMCs experience positive outcomes with e-NAM, it might incentivize others to join. The idea is that APMCs that join in later waves (i.e., in 2018) or APMCs that have yet to join e-NAM benefit from the number of APMCs in their surroundings who are already enrolled. Further, the familiarity of farmers with using the e-NAM platform and their ability to adapt to the online auction system can influence their ability to benefit

from the price discovery mechanism. This heterogeneity across waves implies differential incentives to join and, thus, the possibility of differential treatment effects. Thus, we will use the proportion of already-enrolled APMCs as a proxy for the neighborhood effect of enrolling in e-NAM. We calculate this per APMC per year of interest in our study.

Another potential bias stems from short-sale constraints. APMCs with limited access to institutional credit might be hesitant to join e-NAM due to concerns about pre-harvest financing or potential delays in receiving payment after selling their produce online. The idea is that as the availability of institutional credit decreases, farmers have incentives to search for alternative sources of credit for their immediate requirements. Thus, this could be linked to intermediary dominance, where traditional traders who provide credit might hold more sway over farmers who lack alternative financing options. As existing literature states, middlemen and intermediaries often serve as a financial service, filling the gap in credit access by offering advance credit to the farmers [6]. Hence, we will use the credit-to-deposit ratio of commercial banks per state per year as a proxy for the availability of institutional credit for farmers in a given state. This credit-to-deposit ratio highlights the efficiency of banks in mobilizing their deposits, thus increasing the deployment of credit. We obtain this data from the Reserve Bank of India at the state level for commercial banks from 2005 to 2019 [21].

Further, a district’s standard of living can be a double-edged sword for e-NAM selection. On the positive side, better infrastructure, digital literacy, and financial inclusion (bank access, credit facilities) can enable smooth online transactions, information flow, and efficient produce movement. However, the digital divide, low literacy rates, limited awareness of e-NAM’s benefits, and strong existing local markets can all be barriers to entry. These factors highlight the importance of considering a district’s socio-economic context when studying the impact of e-NAM enrollment. Thus, we also use the percentage of households in the district with low living standards as a proxy for such socioeconomic factors. We use the pre-e-NAM living standards in the ”Standard of Living Index percentages” section of the 2008 District Level Household and Facility Survey (DLHS-3) [27].

2.4 Empirical Strategy

Notably, the three-period roll-out of e-NAM across APMCs (separate waves of enrollment in April 2016, March 2017, and March 2018) suggests using an event study approach. Before considering identification issues, we have a basic form of:

$$p_{iy} = a_i + a_y + \sum_{j=2}^J \beta_j (Pre_j)_{iy} + \sum_{k=1}^K \gamma_k (Post_k)_{iy} + \delta X'_{iy} + \epsilon_{iy} \quad (1)$$

where p_{iy} is the outcome variable indicating a measure of monthly prices for wheat in mandi i at year y ; the time-invariant APMC-level fixed effect a_i controls for possible confounding APMC policies such as the decades-old mandi-sanctioned minimum support price (MSP); APMC-invariant year fixed effect a_y controls for possible year variation like weather; X_{iy} is a vector of observable covariates at the APMC-year level such as number of APMCs already participating in e-NAM per state at that year, the credit deposit ratio of commercial banks per state per year, and the percentage of households with a low standard of living; and ϵ_{iy} is the unobserved error term.

We treat the enrollment of a mandi into e-NAM as our treatment. We denote $eNAM_i$ as an event recording the year y in which e-NAM is adopted in mandi i . The outcome, denoted above as p_{iyt} , takes the year-month level prices for the mandi of interest. Our control group, thus, includes the states in which this event never occurred.

The simplified version of equation (1) is the following:

$$p_{iy} = a_i + a_y + \beta eNAM_i + \delta X'_{iy} + \epsilon_{iy} \quad (2)$$

Equation (2) will be estimated with two specifications of the outcome variable p_{iyt} as mentioned earlier. Both specifications are standardized. The first specification standardizes the average monthly wheat prices of mandis within the same APMC, benchmarked at the APMC average. The second standardizes the average monthly prices of APMCs benchmarked at the state average. Interestingly, for the across-APMC measure, a high deviation indicates that some APMCs within the district experienced significant monthly price changes compared to others, while a low deviation suggests more similar price movements across APMCs. Further, our controls for the already-enrolled APMCs per state, the credit deposit ratio, and the standard of living are *a priori* suspected to correlate with incentives to participate in e-NAM.

Further, the identification of e-NAM's impact relies on the existence of parallel trends in the price differentials (1) across mandis in the same APMC and (2) across APMCs. That is, there should ideally be no deviation before e-NAM's implementation between mandis that will eventually be in e-NAM and mandis that won't. This assumption is of concern because, considering the three waves of enrollment, the timing of joining e-NAM might not be random and pre-e-NAM characteristics of APMCs could be influencing price discovery independent of e-NAM itself. Additionally, there is a concern for the potential spillover effects between waves, creating heterogeneous treatment effects, and spillover across neighboring districts, creating a free-rider issue.

2.5 Sample

This sample consists of all Indian states that have submitted wheat prices (in rupees per quintal) within the span of Jan 2003 to Dec 2019 to AgMarknet [1]. Only Union Territories and the state of Rajasthan were excluded — the former was excluded to keep the governance style consistent and the latter was due to computation errors from the large number of sales.

In order to use districts and APMCs interchangeably in this study, we exclude districts with more than one APMC. Further, in order to analyze mandi and APMC variation, we exclude APMCs that only report wheat prices from one mandi. Thus, our sample for the first specification includes 731 mandis and the second specification includes 588 mandis. We present summary statistics on the specifications and controls by wave and state in Tables 1 and 2, respectively. Notably, narrowing the sample of APMCs to those with more than one mandi has removed the APMCs integrated into e-NAM in wave 3. Thus, Table 1 and the rest of our analysis only pertain to waves 1 and 2.

Column 4 in Table 1 displays the differences in statistics between the untreated mandis and the mandis enrolled into e-NAM on April 2016. Similarly, column 5 displays the differences between the mandis enrolled in e-NAM on April 2016 and March 2017. Regarding the first specification, comparing the mandis who never joined e-NAM to the mandis who were the first to join (column 4), we observe that the average gap between average monthly wheat prices and the APMC average is larger for the e-NAM mandis. Notably, mandis who joined e-NAM first also have a lower percentage of households with a low standard of living as of 2008. Comparing the statistics between the waves (columns 5), we observe that the average price gap is significantly larger for the mandis who joined e-NAM in the second wave, averaging at an average monthly difference of about 20 rupees per quintal of wheat. Further, the percentage of households with a low standard of living, though higher than the mandis who joined e-NAM first, is still lower than the mandis who never joined e-NAM.

For the second specification, we observe a different trend. Contrary to the across-mandi within-APMC trends, we see a consistent decrease in the average price gap, with second-wave APMCs having the lowest average gap of around 172.4 rupees per quintal of wheat. In line with the across-mandi trends, we also observe a vast initial drop in percent of households with low standards of living for APMCs who first joined e-NAM. Moreover, we see that only the APMCs who joined in the first wave belong to states with a greater average credit-to-deposit ratio, while this average ratio is similar between the non-e-NAM and second-wave APMCs. Considering the possibility of initial price volatility given the novelty of the platform in 2016, this may have discouraged APMCs in states with less credit mobility to join. Alternatively, since low credit mobility increases the use of intermediaries as sources of credit, this discrepancy may be due to established physical trader relationships.

Comparing both specifications begs the question of why prices are closer to the state average for APMCs enrolled in e-NAM yet, within mandis in e-NAM, prices become farther away from the APMC average. The law of one price relies on perfect information and the absence of market segmentation. In terms of whether there is a persistence of market segmentation within APMCs, the internal structure of APMCs may be informed by wheat quality variations in the produce sold. The e-NAM platform might be attracting buyers interested in higher-quality produce, leading to higher prices for those specific offerings. This could create a situation where the average price within the APMC diverges as the higher-quality produce finds better deals on e-NAM, while lower-quality produce continues to be traded at lower prices within the APMC itself. However, for these same APMCs, e-NAM may have provided access to information about other state markets, allowing them to align their prices closer to the state average for better competition.

Similarly, Table 2 displays the state-wide statistics. Because joining e-NAM is under the purview of the APMCs, this table illustrates the lack of visible state-wide trends, as we would have hypothesized a link between low state credit-deposit ratios and the standard of living.

Statistics:	Treatment Waves			Between-wave Differences	
	Controls (1)	1st Wave (2)	2nd Wave (3)	$\Delta 1$ (4)	$\Delta 2$ (5)
Specification 1:					
<i>Mandi-APMC Differential</i>					
Mean	70.57	77.75	98.06	7.18	20.31
SD	148.15	166.79	171.11		
<i>State Deposit Ratio</i>					
Mean	65.0	67.1	61.6	2.1	-5.5
SD	14.1	8.70	13.2		
<i>APMC Prev Enrolled Pct</i>					
Mean	0.02	0.14	0.13	0.12	-0.01
SD	0.09	0.31	0.27		
<i>Low Standard of Living Pct</i>					
Mean	51.2	30.4	42.1	20.8	11.7
SD	27.3	12.0	18.5		
No. Mandis	513	124	94		
Specification 2:					
<i>APMC-State Differential</i>					
Mean	219.6	175.2	172.4	44.4	-2.8
SD	249	179.1	141.5		
<i>State Deposit Ratio</i>					
Mean	64.9	71.6	63.9	6.7	-7.7
SD	12.3	9.12	16.0		
<i>APMC Prev Enrolled Pct</i>					
Mean	0.05	0.33	0.23	0.28	-0.1
SD	0.16	0.40	0.31		
<i>Low Standard of Living Pct</i>					
Mean	52.7	32.2	47.2	20.5	15
SD	27.6	13.7	20.7		
No. Mandis	386	112	90		

Columns 4 and 5 show the differences in the means between the control and 1st wave observations, and between the 1st wave and 2nd wave observations, respectively. No wave (i.e. the control) is wave 0. Further, the price differentials are unstandardized, taking only the absolute difference.

Table 1: Summary of Specifications and Controls per e-NAM Wave

State:	Specifications and Controls											
	Spec 1		Spec 2		Deposit	Ratio	APMC Prev	Enrolled Pct	Low Standard of Living Pct	No. APMCs	No. Mandis	
	(1)	(2)	(3)	(4)								(5)
Andhra Pradesh	0.00	0.00			108.2	4.38	0.00	0.00			1	2
Assam	16.48	66.25	163.27	143.66	38.74	2.68	0.00	0.00	67.5	3.94	2	4
Bihar	16.73	26.47			28.11	0.601	0.00	0.00	68.3	18.1	7	16
Chhattisgarh	77.83	26.47	173.63	126.14			0.13	0.20	63.2	21.7	23	101
Haryana	89.05	182.26	181.93	194.91	66.46	7.39	0.14	0.31	28.0	7.76	21	127
Jharkhand	45.60	59.58	136.24	98.55	33.14	2.14	0.00	0.00	51.5	0	4	8
Karnataka	127.90	187.94	294.68	361.34	72.87	3.48	0.00	0.00	61.2	10.7	24	109
Kerala	61.88	154.25	536.41	316.41	63.73	5.34	0.00	0.00	8.56	0.898	4	9
Odisha	15.30	52.06	282.96	256.56	50.07	6.94	0.00	0.00	84.1	8.77	6	13
Punjab	34.18	119.90	175.5	132.63	70.42	8.6	0.00	0.00	12.2	5.08	22	228
Telangana	10.02	43.40	106.46	111.72	101.6	2.34	0.12	0.22	59.1	6.622	7	20
Uttarakhand	74.56	164.0	266.13	208.24	32.91	4.29	0.04	0.09			4	19
West Bengal	25.08	35.32	139.64	128.15	57.7	5.15	0.00	0.00	75.3	5.49	9	33

Columns 1 to 12 are all summaries of specification 1, except for columns 3 and 4. As seen in Table 1, the direction is consistent with both specifications and the magnitude is similar. Thus, specification 1 alone can be informative of state differences.

Table 2: Summary of Specifications and Controls per State

3 Results

3.1 Baseline estimates

Figure 1 shows the dynamic effects of APMCs enrolling in e-NAM over the two waves conditional on the proportion of previously enrolled APMCs, the state credit-to-deposit ratio, and the percentage of households with a low standard of living in the district. Visually, it shows whether the deviation of prices from the benchmark in the mandis or APMCs enrolled in e-NAM differ from those not enrolled. The left panel presents the first specification of prices across mandis in the same APMC and the right panel presents the second specification of prices across APMCs. These panels represent columns (3) and (6) of Table 4. In panel 1, we observe somewhat parallel pretrends, where the mandis that belong to enrolled APMCs deviate similarly to the untreated mandis. To further satisfy the assumption of parallel trends, Table 3 displays the test for the joint nullity of the placebos, which results in largely insignificant pretrends. However, the treatment year presents a discontinuity where the gap between treated and untreated mandis becomes negative, signifying that the deviation within mandis enrolled in e-NAM was less than that of untreated mandis. As for the second specification, there is no significant visual discontinuity near the treatment year.

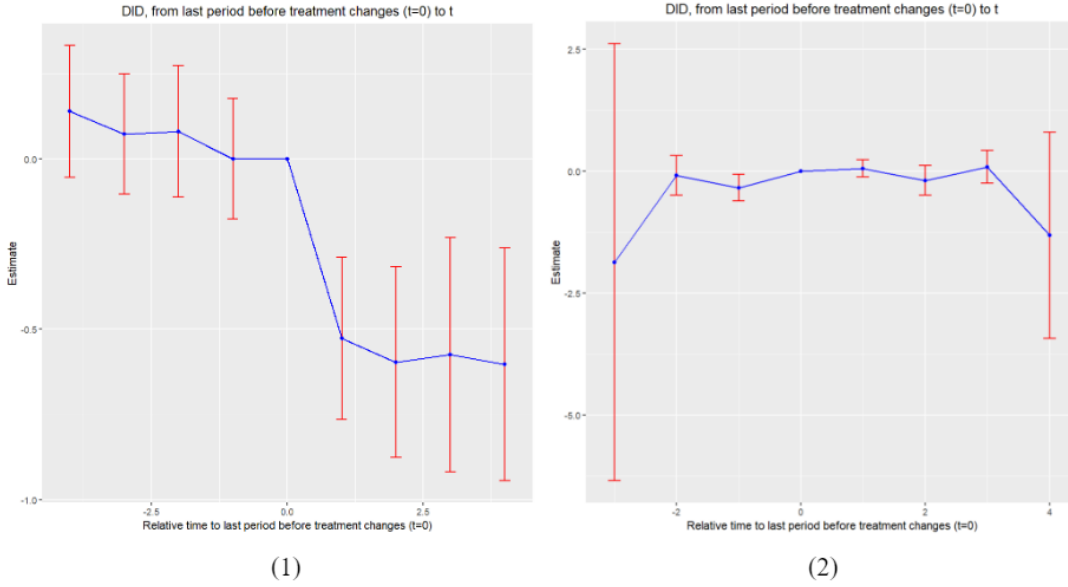


Figure 3: Effect of e-NAM treatment on both specifications

Placebo:	Estimate	SE	N
1	-0.000	0.090	2,125
2	0.081	0.098	2,003
3	0.073	0.090	1,927
4	0.139	0.098	990

Test of joint nullity of the placebos : p -value = 0.556

Table 3: Testing the parallel trends for specification 1

The Chaisemartin and D’Haultfoeuille (2020) estimates of equation (2) are reported in Table 2 [9]. Columns 3 and 6 report the conditional estimates of enrolling in e-NAM’s effect on price convergence. In particular, the first specification aims to assess the baseline effect an APMC enrolling in e-NAM has on the price convergence within its mandis, assuming a uniform impact across all APMCs. In other words, does e-NAM fulfill its price discovery promise of delivering lower? To answer this question, column 3 estimates a significant negative deviation from the APMC average for mandis enrolled in e-NAM. This deviation is estimated to be about 0.6 deviations below the mean. Considering the evidence in the literature supporting the market integration of wheat markets via arbitrage opportunities, we theorized this deviation to at least be positive. It is likely that we are observing price shocks due to the novelty of e-NAM, where e-NAM’s introduction might have increased competition among traders within the APMC. As traders compete to attract farmers selling on the platform, they might offer lower prices than the historical APMC average. We explore this quality variation later by adding a wheat suitability covariate.

Alternatively, we use the second specification to focus on the variation of e-NAM enrollment effects across APMCs, allowing for heterogeneity in the effect of e-NAM not captured in the first specification. It may reflect possible spillover when many APMCs join e-NAM, allowing the few non-enrolled APMCs to be free-riders. Column 6 estimates an insignificant negative deviation from the state average when controlling for the state credit-to-deposit ratio, the standard of living, and the proportion of previously enrolled APMCs in the state. This suggests that on average, e-NAM might not directly lead to price convergence across APMCs relative to the state average. Notably, when just conditioning on the credit-to-deposit ratio, we observe the same significant negative deviation seen in specification 1, of about 0.6 deviations below the state average. Contrary to our hypothesis of upward convergence, both these baseline estimates show a deviation below the specifications’ respective benchmark. This is particularly concerning given the literature showing that wheat prices display long-term price convergence given a change in upstream prices. This negative price gap may be due to our lack of post-periods, likely causing the estimates to fail to smooth over an initial disruption in existing market structures and information flows caused by e-NAM’s pilot programs. Thus the effects of the upstream price change may not be fully realized in our model.

Model:	Dependent Variable					
	Mandi-APMC Differential (1)	(2)	(3)	APMC-State Differential (4)	(5)	(6)
<i>Variables</i>						
eNAM	0.062 (0.061)	-0.030 (0.068)	-0.608 (0.126)	-0.184 (0.076)	-0.602 (0.163)	-0.262 (0.190)
<i>Fixed-effects</i>						
Mandi	Yes	Yes	Yes	No	No	No
APMC	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>						
Deposit ratio	No	Yes	Yes	No	Yes	Yes
APMC prev enrolled	No	No	Yes	No	No	Yes
Standard of living	No	No	Yes	No	No	Yes
<i>Fit statistics</i>						
Observations	8,237	6,007	4,929	2,671	1,810	1,344
F-test p-value	0.031	0.845	0.01	0.00	0.00	0.00

Standard-errors in parentheses. Further, both specifications are standardized. Thus, estimates should be interpreted as the standard deviation of a mandi/APMC's price for mandis/APMCs enrolled in e-NAM. With the effects_equal function of did_multiplegt_dyn, we evaluated the p-value of the F-test on the equality of the effects. This is consistent in later tables.

Table 4: Chaisemartin and D'Haultfoeuille Estimates

3.2 Across-Mandi estimates

Our initial analysis presents a seemingly contradictory finding. While e-NAM is marketed as a platform to increase farmer income, baseline estimates suggest a significant negative price gap across mandis in an APMC enrolled in e-NAM. In line with the assumptions underlying the law of one price, we conjecture that APMCs that join e-NAM don't become integrated markets despite the reduction of transaction costs on the platform — for instance, lower search costs. Rather, it is likely there persists market segmentation within these APMCs based on wheat quality.

There are, unfortunately, ample avenues for e-NAM enrollment to facilitate the two-tiered marketing of wheat based on quality. Because it is common for traditional mandis not to have a robust quality grading system, having an established Quality Assaying lab is a requisite for an APMC to join e-NAM. Pre-e-NAM, it is likely that farmers with higher quality wheat receive the same price as those with lower quality, essentially losing out on potential value. The question is: do we observe a negative price gap due to the APMC average already being inflated from the former mandi system?

While the quality of produce depends on many factors like appearance and texture, we will attempt to proxy quality by a farmer's ability to sell in volume, thus increasing the trader's ability to choose the "best" quality bundles and the perishability of a farmer's produce, which is highly contingent on produce transport. Two covariates we will explore within our first specification are

(1) the mandi's APMC wheat suitability and (2) mandi distance (in km) to the district border. We claim that including wheat suitability, proxied by the APMC's output density data gathered from GAEZ [29], allows us to capture the potential influence of bulk selling and its impact on farmer leverage within the e-NAM platform. This can provide valuable insights into how e-NAM enrollment affects price convergence for crops grown in APMCs with varying production capacities. Similarly, including the marginal effect of distance can reflect the differentiated gains due to farmers' market access. The proximity of a mandi to its district boundary can indicate not only the mandi's own transportation costs but also an out-of-district intermediary's willingness to travel [8]. The idea is that a lower chance of perishability increases the product's allure.

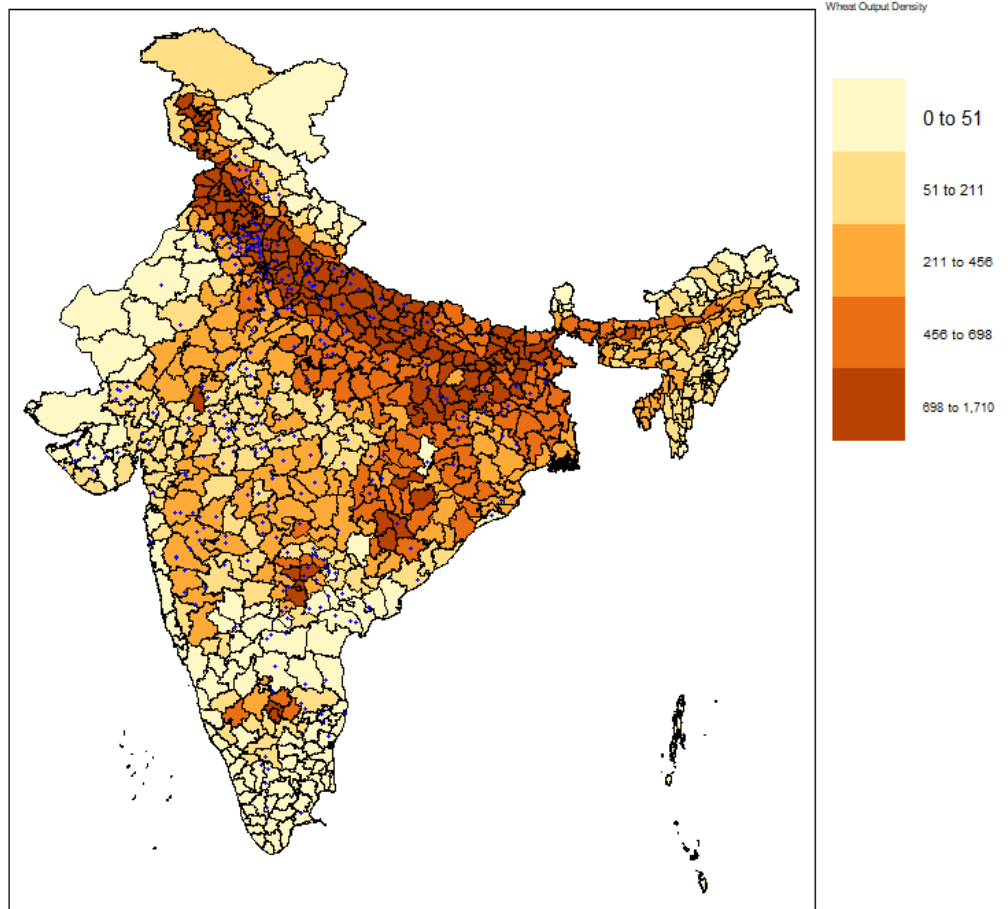


Figure 4: Mandi Locations and APMC-level Wheat Output Density Map

Beyond the relationship with quality variation, APMCs with higher crop suitability will likely produce larger volumes of the specific crop under study. This aligns with the growing trend of bulk buying in the food industry, where large retail chains and supermarkets prefer to purchase in bulk for efficiency and cost savings (as highlighted by the dominance of powerful buyers in the agricultural

value chain) [28]. Selling in bulk through e-NAM allows farmers in these APMCs to share logistical costs, making the platform even more attractive than local markets. Furthermore, farmers can potentially present a more unified front on e-NAM by aggregating their produce through higher yields. This strengthens their bargaining power compared to selling smaller quantities individually, particularly in a context characterized by limited options for small farms and a reliance on informal agreements with local intermediaries who often hold a stronger bargaining position.

	Dependent Variable				
	e-NAM	Mandi-APMC Differential			
Model:	(1)	(2)	(3)	(4)	(5)
<i>Variables</i>					
eNAM			-0.609 (0.126)	-1.828 (1.644)	-0.675 (0.182)
Distance to border	0.000 (0.000)				
Mean output density		0.000 (0.000)			
Deposit ratio	0.000 (0.000)	0.000 (0.000)			
APMC prev enrolled	1.441 (0.193)	0.871 (0.012)			
<i>Fixed-effects</i>					
Mandi	Yes	Yes	Yes	Yes	Yes
APMC	Yes	Yes	Yes	Yes	Yes
Year	No	Yes	Yes	Yes	Yes
<i>Baseline Controls</i>					
Deposit ratio	Yes	Yes	Yes	Yes	Yes
APMC prev enrolled	Yes	Yes	Yes	Yes	Yes
Standard of living	No	No	Yes	Yes	Yes
<i>Other Controls</i>					
Distance to border	No	No	No	Yes	No
Mean output density	No	No	No	No	Yes
<i>Fit statistics</i>					
Observations	4,431	4,431	4,929	554	1,231
F-test p-value			0.031	0.00	0.00
<i>Standard-errors in parentheses. e-NAM enrollment is not standardized while Mandi APMC Differential is. Further, columns 1 and 2 are clustered at the state level. The standard of living measure was also removed from columns 1 and 2 due to collinearity with distance.</i>					

Table 5: Chaisemartin and D’Haultfoeuille Estimates with Added Covariates for Specification 1

Figure 4 displays the variation in wheat output density across the districts in India alongside the location of 283 mandis in e-NAM [23] that were geocoded using Google’s API. While the 283 mandis are not a representative sample and were narrowed from what Google could capture, our use of a suitability measure is bolstered by the concentration of mandis in areas with an output density in the highest quintile.

Table 5 provides empirical evidence for the relevancy of measures for suitability and distance. Columns 1 and 2 show that, controlling for our other baseline variables, suitability and distance have no individual predictive ability of e-NAM enrollment. However, column 5, when controlling for suitability, estimates a significant negative deviation from the APMC average consistent with and larger than our baseline estimate for specification 1 in Table 4. Column 4, on the other hand, estimates an insignificant negative deviation when controlling for distance. While our computations failed to regress both suitability and distance in one model, we can conjecture that a mandi’s distance to the border explains much of the negative deviation.

We initially hypothesized that distance to the border might be a proxy for perishability. Mandis closer to the border might have a lower chance of perishability, potentially attracting out-of-district intermediaries willing to travel for the produce, leading to a less negative price deviation from the APMC average. However, the estimates from Table 5 suggest the possibility of an APMC’s selection into e-NAM due to distance. It’s possible that APMCs that decided to enroll in e-NAM were already located in areas closer to the border (thus, with lower perishability risk) in the first place. These factors might independently lead to less negative price gaps within those APMCs, though column 1 estimates a near zero direct impact of distance on e-NAM enrollment.

3.3 Wave Heterogeneity

The consistent negative deviation in the first specification suggests that, on average, mandis in APMCs enrolled in e-NAM might be receiving lower prices compared to the APMC average price. We further explain this deviation by analyzing the heterogeneous treatment effect across different e-NAM periods. Rather than a change in the direction of deviation, we anticipate that analyzing group-specific trends minimizes the magnitude of negative deviation.

There are various reasons for the second wave APMCs to experience greater benefits than the first. Firstly, e-NAM is a platform that relies on network effects. As more APMCs join, the platform becomes more valuable for everyone involved. Later-enrolling APMCs might benefit more from an already established network of buyers and sellers, potentially leading to a stronger positive effect on price convergence. Further, there might be a learning curve associated with using e-NAM effectively. Early adopters might have faced initial challenges in utilizing the platform to its full potential. Later waves could benefit from the experience of earlier adopters, leading to a faster and more pronounced impact on price convergence. While we try to address this by controlling for the proportion of APMCs already enrolled in e-NAM, perhaps there is a better way to capture this effect.

Thus, to account for wave heterogeneity, we estimate group-specific linear trends by regressing separately for each enrollment wave. Notably, the only variable to significantly differ across treatment periods is the credit-to-deposit ratio’s impact on the price gap. Comparing these results to Table 1, our summary statistics by wave, we do not see any compelling differences across treatment periods.

Model:	Mandi-APMC Differential		
	Controls (1)	1st Wave (2)	2nd Wave (3)
<i>Variables</i>			
Mean output density	0.000 (0.000)	0.0004 (0.0001)	0.0004 (0.0001)
Deposit ratio	0.0097 (0.0022)	-0.0006 (0.003)	0.009 (0.003)
APMC prev enrolled		-0.243 (0.084)	-0.059 (0.109)
Standard of Living	-0.0006 (0.0007)	0.004 (0.003)	-0.002 (0.004)
<i>Fixed-effects</i>			
Mandi	Yes	Yes	Yes
<i>Baseline Controls</i>			
Deposit ratio	Yes	Yes	Yes
APMC prev enrolled	Yes	Yes	Yes
Standard of living	Yes	Yes	Yes
<i>Other Controls</i>			
Distance to border	No	No	No
Mean output density	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	2,669	1,504	839

Standard-errors in parentheses. Distance was not included due to collinearity with the standard of living measure. Further, year fixed effects were not included since no variables are time-variant.

Table 6: Fixed Effects Estimates Across Wave Assignments

4 Threats to Validity

Many APMCs have missing price data for certain months, creating an unbalanced panel structure at the year-month level. This missing data could be related to the treatment variable (e-NAM enrollment) as self-reporting accurate transaction data measures faith and use of e-NAM.

Further, using APMC enrollment into e-NAM as a proxy for farmer participation in e-NAM introduces an additional layer of uncertainty. While enrollment indicates access to the platform, it doesn't necessarily reflect the extent to which individual farmers actively utilize it. This is reflected in the negative baseline estimates for specification 1, where it was found that there is a significant negative price deviation across mandis in an APMC enrolled in e-NAM.

Additionally, there is concern for spatial correlation. Prices in nearby APMCs might be more similar to those in distant APMCs. This could be due to shared transportation infrastructure, sim-

ilar agricultural products, or information diffusion between neighboring markets. In contrast, there could be spatial heterogeneity as the effect of e-NAM on price convergence might vary depending on the connectivity. That is, the impact might be stronger in districts with a higher density of APMCs or better connectivity to major markets.

In this section, we will describe several robustness checks to mitigate some of these threats.

4.1 Unbalanced panel implications on self-reporting

We attempt to mitigate the potential bias that a mandi’s missing price data might introduce on the relationship between e-NAM enrollment and price convergence.

Column 1 of Table 7 estimates whether the suitability or distance of a mandi in e-NAM is associated with the mandi’s lack of price data. We previously anticipated that a mandi is less likely to report data if it has low suitability or a larger distance. A lower suitability might indicate lower production volume, meaning the mandi might have fewer transactions and less incentive to invest time and resources in reporting data on e-NAM. Additionally, the benefits of increased visibility and access to a wider market through e-NAM might be limited. This could be because they have fewer resources to leverage the platform effectively for price discovery or finding better deals. A similar argument can be made for mandis with larger transport costs. Yet, seeing virtually no predictive power, we estimate control for a mandi’s missing data in column 3.

Comparing the original model in column 2 and our new model in column 3, we observe that controlling for missing data makes our negative deviation insignificant. However, the direction of deviation is consistent. Hence, we can infer that, firstly, there is no statistically significant association between our covariates and missing price data and, secondly, missing price data does not seem to bias our results.

Model:	Dependent Variable		
	Pct of Missing Price Data (1)	Mandi-APMC (2)	Differential (3)
<i>Variables</i>			
eNAM		-0.675 (0.182)	-0.197 (0.173)
Mean output density X treated	0.000 (0.000)		
Distance to border X treated	0.000 (0.000)		
<i>Fixed-effects</i>			
Mandi	Yes	Yes	Yes
APMC	Yes	Yes	Yes
Year	Yes	Yes	Yes
<i>Controls</i>			
Pct Missing Price Data	No	No	Yes
Deposit ratio	Yes	Yes	Yes
APMC prev enrolled	Yes	Yes	Yes
Standard of Living	Yes	Yes	Yes
Mean output density	Yes	Yes	Yes
Distance to border	Yes	No	No
<i>Fit statistics</i>			
Observations	1,408	821	821
F-test p-value	0.00	0.031	0.00

Standard errors in parentheses, clustered by district for column 1. Further, estimates of zero are rounded from significant figures less than 10^{-14} . Standard of living, suitability, and distance were also removed from column 1 due to collinearity

Table 7: Robustness test for the endogeneity of nonreponse

4.2 Standardization of across-mandi within-APMC price differentials

Our analysis employs standardized price differentials across mandis within the same APMC, rather than absolute price differences. This choice is crucial because it addresses a key assumption underlying our event study approach — the parallel trends assumption.

Figure 5 depicts the event study results using the standardized prices in panel 1 and not standardized prices in panel 2. As panel 2 illustrates, the absolute price differences between mandis in APMCs enrolled in e-NAM and not exhibit significant positive deviation even before e-NAM enrollment. Standardizing allows us to focus on the relative price difference between mandis within an APMC, rather than the absolute monetary value. Further, as seen in Table 8, standardization yields more significant estimates and a direction of where the deviation from the average is occurring. This becomes particularly important in light of the negative deviation we observe in our

analysis. This negative deviation suggests that, on average, farmers within e-NAM APMCs are receiving lower prices compared to the APMC average. This is a critical finding that wouldn't be readily apparent if we were solely relying on absolute differences.

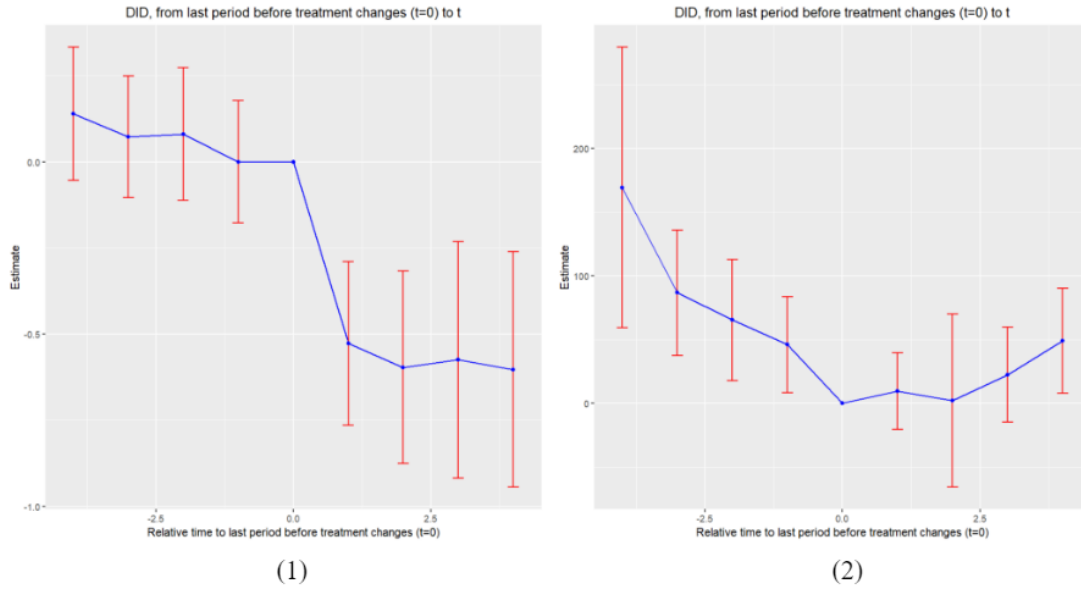


Figure 5: Impact of standardization on the pretrends for Specification 1

Model:	Mandi-APMC Price Differential					
	Standardized			Not Standardized		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
eNAM	0.063 (0.067)	-0.037 (0.071)	-0.622 (0.13)	25.13 (13.82)	20.56 (17.41)	24.98 (18.36)
<i>Fixed-effects</i>						
APMC	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>						
Deposit ratio	No	Yes	Yes	No	Yes	Yes
Standard of living	No	No	Yes	No	No	Yes
APMC prev enrolled	No	No	Yes	No	No	Yes
<i>Fit statistics</i>						
Observations	7,601	5,791	4,785	7,602	5,846	4,785
F-test p-value	0.031	0.88	0.01	0.009	0.069	0.012

Standard-errors in parentheses.

Table 8: Alternatives to Specification 1

4.3 An alternative benchmark for across-APMC price differentials

Using an alternative benchmark addresses the simultaneous determination of state average prices, whereby we include our APMC of interest's average value to calculate the state average. Hence, we risk the fluctuations in district price significantly informing fluctuations in state averages, especially considering the states in Table 2 that weigh each district more due to scarce district data. Further, state-level factors, such as overall agricultural production or government policies, might influence both e-NAM enrollment decisions by APMCs and the state average price. This creates a potential correlation between e-NAM enrollment and the benchmark price, leading to biased estimates of e-NAM's true impact on price convergence.

While we originally used state averages to observe whether e-NAM promotes convergence within a specific geographic context, in this section, we use the wholesale price inflation index price as a benchmark for inflation-adjusted prices. This allows us to analyze price convergence while accounting for general price increases over time. This is important because nominal price convergence might not necessarily reflect real price convergence if inflation is a significant factor. Consistent with the observations of specification 1, we observe a negative deviation from our respective benchmarks in columns 3 and 6, albeit both insignificant. Hence, our choice of benchmarks is inconsequential to our overall conclusions.

Benchmark: Model:	Across-APMC Price Differential					
	State	Monthly	Average		Yearly WPI	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
eNAM	-0.184 (0.076)	-0.602 (0.163)	-0.262 (0.190)	-0.703 (0.944)	-2.772 (2.545)	-1.490 (3.258)
<i>Fixed-effects</i>						
APMC	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
<i>Controls</i>						
Deposit ratio	No	No	Yes	No	No	Yes
APMC prev enrolled	No	No	Yes	No	No	Yes
Standard of living	No	Yes	Yes	No	Yes	Yes
<i>Fit statistics</i>						
Observations	2,671	1,810	1,344	2,762	1,810	1,344
F-test p-value	0.00	0.00	0.00	0.00	0.00	0.00

Standard-errors in parentheses. Further, both specifications are standardized. Thus, estimates should be interpreted as the standard deviation of an APMC's price for APMCs enrolled in e-NAM.

Table 9: Alternatives to Specification 2

5 Conclusion

This paper documents the avenues through which e-NAM, an electronic trading platform for agricultural products in India, can affect price convergence across markets. Our initial hypothesis predicted positive convergence, but the results revealed a surprising negative deviation. On average, mandis within e-NAM experience average monthly prices 0.6 standard deviations below their APMC average monthly wheat price with inconclusive evidence of e-NAM's impact on across-APMC price convergence.

We explored constraints to farmers utilizing e-NAM fully, including their access to agricultural credit, prior expectations from neighboring APMCs joining, standard of living and mandi-specific factors such as wheat suitability and proximity to the district border. However, the significant results from these factors were still consistent with the prices following a downward trend.

Further research is needed to fully understand the negative deviation. Potential explanations include quality segmentation within markets, where higher quality produce benefits from e-NAM while lower quality produce remains at lower prices in traditional systems. Additionally, limited participation in e-NAM by some farmers or markets could create a price gap. Strategic manipulation by local traders targeting farmers outside the platform is another possibility.

In particular, specification 1's inconsistency with previous literature on wheat prices supports this incomplete market integration, where farmers who are not actively using e-NAM might still rely on traditional market structures and local power dynamics. This raises the question of how

e-NAM usage varies among farmers and how it influences their market outcomes. A key limitation of our current analysis is the difficulty in directly observing farmer usage of e-NAM from the mandi-district data we use. Our data provides valuable insights into market trends and APMC enrollment, but it doesn't capture individual farmer participation on the platform. Given these limitations, we acknowledge the need for alternative approaches to assess farmer usage of e-NAM. Future research incorporating data on individual farmer engagement with the platform would provide a more comprehensive picture of how e-NAM impacts market integration and price convergence for all farmers.

More concerning, perhaps, is specification 2's insignificance. In its most extreme interpretation, it suggests a complete pass-through of price changes, potentially reflecting the continued dominance of local factors or the limited geographical reach of e-NAM's price discovery mechanism. The lack of significant price dispersion across APMCs for wheat, but its existence within individual APMCs, suggests that transport costs and distance might play a role at the inter-APMC level. For future studies, this can be explored via a proxy for transaction costs, such as the freight rates for wheat, or data on perishability, such as storage locations and co-location of processing units.

Finally, given the robust literature on price convergence and the integration of wheat markets, our results are limited by our short-term transaction data. Potential studies should expand their time horizon to smooth for possible initial price volatility, a concern we saw in our first specification. In the long run, e-NAM participation could reflect significant price convergence towards a higher producer price.

Despite our negative findings, this knowledge can inform strategies to ensure all farmers benefit from e-NAM participation, ultimately promoting fair and equitable prices in the Indian agricultural market. While e-NAM's impact on price convergence appears more complex than initially anticipated, further exploration is crucial to maximizing the platform's effectiveness.

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